

Fixed couple ménage rows

Exact differences, concavity, uniform expansions and inverse enclosures

Report 221

4 October 2026

Abstract

We prove the adjacent-difference and row-unimodality conjectures displayed in OEIS A332709. A path-rook polynomial identity expresses every adjacent difference in terms of A127548, with a separate zero at the even central step. An explicit injection proves discrete concavity and determines every equality case, including the four-entry maximum plateau in even rows with $n \geq 6$ and strict log-concavity in odd rows. An exact decomposition about the row mean yields an expansion to every fixed order, uniformly over all allowed first images. It separates the boundary layers from the interior, gives the fourth-column expansion A258667 through relative order ten, and yields an exact total-variation identity in every row and its sharp asymptotics. Finally, we distinguish exact-range recovery by a smooth asymptotic model from a threshold inverse, proving a two-candidate threshold enclosure with existential constants. The latter is not a certified numerical algorithm. Classical rook theory, the published fixed-couple factorization, ordinary ménage asymptotics and general inverse methods are credited. Historical checking is bounded, not an exhaustive novelty claim. A portable exact-arithmetic implementation and separate numerical diagnostics accompany the proofs.

[write] **Status and trust boundary, added 7 October 2026, batch 113.** This is bundle Report 221 of an external AI-assisted research session, filed in ProveIt as the single-source research report **a332709-fixed-couple-menage** (provenance, the sources as the write read them, what was checked, relation to the repository, the collected non-claims and the reading conventions are in Section 1.2). It is unrefereed and not formalized. Its author line and PDF author field read “Report 221”: it names no person or tool. *What rests on what.* Every theorem has a conventional proof in the text: rook inclusion–exclusion and the Shevelev–Moses factorization, a polynomial identity for path matchings, explicit injections, an exact telescoping identity, factorial tail bounds, the classical Kaplansky–Riordan expansion with the Wyman–Moser rounding formula, and the mean value theorem. No proof uses a computation; the exact checks corroborate finite identities and the floating diagnostics are illustrations. The write adds Remarks 9.2 and 11.1 and dated notes; no statement, proof or number of the source was changed.

1 The problem and the conclusions

For $n \geq 3$ and $3 \leq k \leq n$, let $T(n, k)$ count the permutations π of $\{1, \dots, n\}$ such that

$$\pi(i) \notin \{i, i+1 \pmod{n}\} \quad (1 \leq i \leq n), \quad \pi(1) = k.$$

The residue 0 is represented by n . These are the fixed-first-image ménage counts in A332709 [1]. We use the reindexing

$$A(n, s) = T(n, s+2), \quad 1 \leq s \leq n-2, \quad h = \left\lfloor \frac{n-1}{2} \right\rfloor, \quad q = \min\{s, n-1-s\}. \quad (1)$$

Thus q is the distance in column indices from the nearer endpoint, counting the endpoint as distance one. The sum of a row is the ordinary circular ménage number U_n . The columns $s = 1, 2, 3, 4, 5$ are respectively A258664, A258665, A258666, A258667 and A258673. In the seating interpretation their clockwise chair separations are $2s + 1$.

The displayed A332709 entry, revision #33 dated 1 February 2021, still labels row unimodality and the adjacent-difference formula as conjectures. The natural domain of the latter includes $4 \leq k \leq n$: the predecessor must itself be a column. We prove

$$T(n, k) - T(n, k - 1) = A127548(n - 2k + 4) \quad (4 \leq k \leq n, n \geq 2k - 3).$$

The even central difference is zero; it is not obtained by substituting zero into A127548. Shevelev and Moses had already observed the row-unimodality pattern in the published 2016 paper [5]. Their factorization is the starting point here.

The stronger row-shape conclusions are exact for every n . The asymptotic results are fixed-order statements as $n \rightarrow \infty$, with the relative row expansion uniform in s . The inverse results concern a fixed column s , not a column varying arbitrarily with n . These quantifiers will be kept separate.

1.1 A fourth-column consequence

For $a_n = A258667(n) = A(n, 4)$ when $n \geq 6$,

$$\begin{aligned} \frac{(n-2)a_n}{U_n} &= 1 + \frac{2}{n^4} + \frac{16}{n^5} + \frac{90}{n^6} + \frac{456}{n^7} + \frac{2296}{n^8} \\ &\quad + \frac{12051}{n^9} + \frac{67193}{n^{10}} + O(n^{-11}). \end{aligned} \tag{2}$$

Thus the fourth-column count and the row average first differ at the fourth relative order. The leading equivalence posted in A258667 and proved in the linked Lean source remains correct [4, 10]. Equation (2) supplies column-specific higher-order information while preserving that leading equivalence.

1.2 Provenance, sources and reading conventions

[write] **Added 7 October 2026, batch 113.** *Provenance.* This report is Report 221 of the session bundle that ProveIt's docs/incoming intake processed as batch 113: the archive `Report221.zip` (475,232 bytes, 16 files, no wrapper directory), arrival commit `60f54ea06`, placed in `a4186a946`. The text is the delivered `Report221.tex` (591 lines, 16 pages, dated 4 October 2026), shipped as `article.tex`. The programs `menage.py`, `check_exact.py` and `diagnostics.py` are shipped under `code/` with the delivered root `build.py`; the five receipts (the four byte-equal exact receipts of the normal, optimized and low-digit-cap runs, and the diagnostic receipt), the requirements and `source_pins.json` under `data/`; `README.md` and `SOURCES.md` were delivered at the root, and the delivered `README.md` was staged as the report `README` and is replaced by it. Not shipped, and retrievable from the arrival commit: the delivered `Report221.pdf` and the pure checksum manifest `MANIFEST.sha256` (15 entries, verified at the write). The package carries no “prepared for private review” line (the `README`'s sentence on private review reports is an exclusion) and no personal data, and records no ProveIt commit; its source ledger pins the Git blobs of two repository files, which are the current ones (below). It is a single-source report, so no merge choice was made. The write prefixed the 60 delivered labels with `fcm:` and updated the references to them (`36 \eqref`, `12 \ref`); it added this subsection, the notes marked [write] and Remarks 9.2 and 11.1, each the last statement of its section, and inserted nothing before a delivered display, statement or table, so every theorem, section, equation and table of the source keeps its delivered number.

The sources, as the write read them (7 October 2026): OEIS A332709, A127548, A000179, A258664, A258665, A258666, A258667 and A258673 in the internal format, with the b-files of A332709, A127548, A000179 and

A258667 (Remark 11.1); the Lean file of the A258667 comment, at the current main branch of its repository (the source’s pinned SHA-256 is that of the current file with one trailing newline added; the statements it quotes, `target_theorem_0`, `seq_equiv` and `asympt_term_equiv`, are `IsEquivalent` statements, as it says); the repository volumes `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex` and `Analysis/Transseries/docs/series-and-transseries/Combinatorial_Transseries_Inverses/Combinatorial_Transseries_Inverses.tex`, whose current Git blobs are the ones the source pins. The first volume’s title is “Transseries / The polynomial–logarithmic calculus, series reversal at infinity, and the inversion of rapidly growing functions”; the source’s `SOURCES.md` and bibliography shorten its subtitle to “... and inversion”. Not read by the write: the cited literature (Shevelev–Moses, Wyman–Moser, Kaplansky–Riordan, Kagey, Gu–Zhao, Knuth’s note); the source’s account of it (Section 11 and `SOURCES.md`) is reported, not confirmed. The source records that it did not obtain the full texts of Gu–Zhao and of Kagey’s published version.

What was checked at intake. The batch-113 dossier read the manuscript, found no demonstrably wrong claim, confirmed that the two A332709 conjectures are still displayed, and reran the delivered suite on a copy (the four exact receipts are byte-equal and are staged as delivered records). *What the write checked.* It read every proof and found no error. With its own code (Python 3.14.4, exact integers and fractions; SymPy 1.14.0; mpmath 1.3.0 at 400 digits), not importing the delivered programs, it

- computed $T(n, k)$ from the cofactor formula (5) for $n \leq 160$; it equals all 1275 terms of the A332709 b-file (rows $3 \leq n \leq 52$), the entry’s own double-sum formula ($n \leq 20$), and a brute-force count over all permutations ($n \leq 9$); the row sums equal the A000179 b-file, L_m the A127548 b-file, and column four the A258667 terms;
- checked, for every $n \leq 160$, Theorem 3.2 at every column, the complete classification of Theorem 4.2 (maxima, concavity equalities, log-concavity equalities), the three identities of Theorem 5.1, and the exact total-variation identity and sign pattern of Theorem 7.2; and (11), (31), (28), the line recurrence, Lemma 4.1, the growth bounds and (32) of Lemma 7.1, and $U_{m-1} \leq L_m$ in the same range;
- derived the coefficients of Table 1 by its own route (note after the table), confirmed them by exact residuals, and the total-variation coefficients of (33);
- checked the Wyman–Moser rounding (23) for $2 \leq m \leq 160$, the expansion (38) and the inverse start (40) at exact range values (residuals of the stated orders), and the stronger reading of the A258667 conjecture (Remark 11.1).

It reran, on a fresh copy of the shipped programs, `check_exact.py` in its four modes (normal, `-0`, digit cap 640, both) and `diagnostics.py`: every output equals the shipped receipt after removing carriage returns. The builder was not run. Floating computations are observations, not certificates.

Relation to the repository. Before batch 113 no file of the repository named A332709, A127548, A258667 or the ménage numbers A000179. The source credits two repository volumes for its inverse distinctions; they are compared statement by statement in Remark 9.2. No result of another report is shared, so no reciprocal note is proposed. No Lean or Rocq file treats these sequences; no statement of this report is formalized (the Lean file above is external and concerns the leading equivalent of A258667 only), and its place in the collection confers no formal status.

[write] Independent check, 7 October 2026. “Whose current Git blobs are the ones the source pins” was true at the write (b353bba89). Later the same day d92db8d06 added two dated repairs to Appendix V (the Fubini pole lattice) of the first volume, whose blob is now f6f91ca842881a9d63a518120f01e36aa91a1fff. No labelled statement was added or changed, and the statements compared in Remark 9.2 are unchanged. The second volume’s blob is still the pinned one.

[write] What is not claimed, collected from the source. The rook inclusion–exclusion, the Shevelev–Moses fixed-couple factorization and their observation of row unimodality, Kagey’s prefix framework, the Kaplansky–Riordan expansion, the Wyman–Moser rounding formula and the repository’s inverse distinctions are prior; no global historical-novelty or first-proof claim; the publisher texts of Kagey and of Gu–Zhao were not read, so non-overlap with them is not claimed. The asymptotic statements are fixed order; the inverse results concern a fixed column; the inverse constants and onsets are existential, with no certified threshold algorithm and no single-ceiling formula; the diagnostics are floating computations; the Lean proof was read,

not rerun. *The write adds:* its checks are exact or floating computations; its transseries remark claims no novelty for any inversion.

Reading conventions (letters the source uses in more than one sense, with the tempting false readings; no symbol was renamed).

Symbol	Meanings
T, A	$T(n, k)$ the triangle (A332709), $3 \leq k \leq n$; $A(n, s) = T(n, s + 2)$. <i>False reading:</i> $A(n, s)$ is not an OEIS A-number.
$L, \mathcal{L}$	L_m the auxiliary counts (A127548, $L_0 = 1$); $\mathcal{L}(z)$ their generating function; $\mathcal{L}(S)$ the law of S in (33). <i>False reading:</i> the zero central difference is not L_0 .
$U, \mathcal{U}, u$	U_m the ménage numbers with the signed values $U_0, U_1, U_2 = 1, -1, 0$; $\mathcal{U}(z)$; $u(n) = U_n / (e^{-2} n!)$.
$R, \mathcal{R}$	$R_\ell(z)$ the path matching polynomials; $\mathcal{R}(n, s)$ the relative row profile (16).
$F, \mathcal{F}, \Phi$	$\mathcal{F}(z) = \sum r! z^r$; F_j the polynomials in the proof of Lemma 3.1; $F_{s,M}$ the inverse model and $F_\pm$ its envelopes; Φ_N the rook functional.
S, E	$S_M(v)$ the ménage amplitude (21); $S = \pi(1) - 2$ a random column; $\mathcal{S}_m$ permutation sets; $E_M(v)$ the line amplitude.
h, q, s, k	$h = \lfloor (n-1)/2 \rfloor$; $q = \min\{s, n-1-s\}$ the distance from the nearer endpoint; s and $k = s+2$ column indices.
m, N, M	m the difference parameter $n-2k+4$ and an index; N the degree in Φ_N , an onset in Theorem 9.1 and in $N_s(y)$ the threshold; M an expansion order.
X, ℓ, W	$X, \ell = \log X$ the Lambert scale (39); ℓ also a path length; W the Lambert function and $W_{p,q}$ the polynomials in Lemma 3.1.
x, c, C	$x = 1/n$ in Table 1 and a real variable in Sections 8–9; $c = \frac{1}{2} \log(2\pi) - 2$ and $c_r(q)$ the coefficients; C a constant.
a_n, D, J, K	$a_n = A(n, s)$ a fixed column (the fourth in (2)); $D(n, s)$ the deviation in Theorem 7.2; J a truncation depth; K an expansion order.

2 Classical rook machinery and conventions

2.1 Path polynomials and the fixed-couple factorization

Let $R_\ell(z)$ be the matching polynomial of a path with ℓ edges:

$$R_\ell(z) = \sum_{j=0}^{\lfloor (\ell+1)/2 \rfloor} \binom{\ell+1-j}{j} z^j, \quad \ell \geq 0. \quad (3)$$

A matching is a set of nonconsecutive edge positions; subtracting the mandatory gaps gives the displayed binomial coefficient. Deleting the final edge, according to whether it is used, gives

$$R_0 = 1, \quad R_1 = 1 + z, \quad R_\ell = R_{\ell-1} + zR_{\ell-2} \quad (\ell \geq 2).$$

For a polynomial p of degree at most N , set

$$\Phi_N(p) = \sum_j (-1)^j [z^j] p(z) (N-j)!. \quad (4)$$

The rook inclusion-exclusion formula counts allowed permutations by applying Φ_N to the rook polynomial of the forbidden board. Indeed, a choice of j nonattacking forbidden squares fixes j row-column assignments and leaves $(N-j)!$ completions; all other choices are inconsistent and contribute zero.

The forbidden row-column incidence graph for a circular ménage permutation is a cycle with $2n$ edges. Fixing $\pi(1) = k$ deletes row vertex 1 and column vertex k . These vertices are nonadjacent when $3 \leq k \leq n$. The remaining graph is the disjoint union of paths with $2k - 5$ and $2n - 2k + 1$ edges. Rook polynomials multiply over components, so

$$T(n, k) = \Phi_{n-1}(R_{2k-5}R_{2n-2k+1}), \quad A(n, s) = \Phi_{n-1}(R_{2s-1}R_{2n-2s-3}). \quad (5)$$

This is the known Shevelev–Moses factorization, equations (30)–(31) of [5]; the path-polynomial method also occurs in the general prefix framework of Kagey [8]. Neither the factorization nor rook inclusion-exclusion is claimed as new.

Swapping the two factors proves reflection:

$$T(n, k) = T(n, n + 3 - k), \quad A(n, s) = A(n, n - 1 - s). \quad (6)$$

Every entry is positive: the cyclic shift $\pi(i) = i + k - 1 \pmod{n}$ is admissible and has first image k . Partitioning all ménage permutations by their first image gives

$$\sum_{s=1}^{n-2} A(n, s) = U_n. \quad (7)$$

2.2 The auxiliary sequence

For $m \geq 1$, let

$$L_m = \Phi_m(R_{2m-2}) = \sum_{j=0}^{m-1} (-1)^j \binom{2m-1-j}{j} (m-j)!. \quad (8)$$

It counts permutations σ of $[m]$ satisfying $\sigma(i) \notin \{i, i+1\}$ for $1 \leq i < m$, with the last row unrestricted. The forbidden graph is a path with $2m - 2$ edges and an isolated row vertex. Consequently

$$0 \leq L_m \leq m!, \quad (L_1, L_2, L_3, L_4, L_5) = (1, 0, 1, 4, 19). \quad (9)$$

Writing $\mathcal{F}(z) = \sum_{r \geq 0} r! z^r$ as a *formal* series, the coefficient of z^m in $\mathcal{F}(z/(1+z)^2)$ is, for $m \geq 1$,

$$\sum_{r=1}^m (-1)^{m-r} r! \binom{m+r-1}{m-r} = L_m.$$

The substitution $j = m - r$ proves the equality. Thus L_m is A127548, whose formal constant term is $L_0 = 1$ [2]. No analytic convergence is being asserted.

For generating-function algebra only, use the signed Touchard convention

$$U_0 = 1, \quad U_1 = -1, \quad U_2 = 0. \quad (10)$$

For $n \geq 3$ these are the usual positive counts. The known ordinary generating function [3, 7] is

$$\mathcal{U}(z) = \sum_{m \geq 0} U_m z^m = \frac{1-z}{1+z} \mathcal{F}\left(\frac{z}{(1+z)^2}\right).$$

Multiplication by $(1+z)/(1-z)$ gives the exact, classical relation

$$L_m = U_m + 2 \sum_{r=0}^{m-1} U_r \quad (m \geq 1). \quad (11)$$

For $m \geq 2$ the small signed terms cancel, and this can be written $L_m = U_m + 2 \sum_{r=3}^{m-1} U_r$ with the sum empty when appropriate. Using $U_1 = 0$ in (11) is incorrect, already at $m = 2$.

3 Exact adjacent differences

Lemma 3.1. *For integers $a \geq 0$ and $b \geq a + 2$,*

$$R_{a+2}R_b - R_aR_{b+2} = (-z)^{a+2}R_{b-a-2}. \quad (12)$$

Proof. Define $F_0 = 0$, $F_1 = 1$ and $F_{j+1} = F_j + zF_{j-1}$. Then $R_r = F_{r+2}$. For $q \geq p \geq 0$, let

$$W_{p,q} = F_{p+1}F_q - F_pF_{q+1}.$$

The recurrence gives $W_{p,q} = -zW_{p-1,q-1}$ for $p \geq 1$, while $W_{0,q-p} = F_{q-p}$. Hence $W_{p,q} = (-z)^p F_{q-p}$. On using the recurrence once, the left side of (12) becomes

$$R_{a+1}R_b - R_aR_{b+1} = W_{a+2,b+2} = (-z)^{a+2}R_{b-a-2}.$$

□

Theorem 3.2 (All adjacent differences). *For $4 \leq k \leq n$, put $m = n - 2k + 4$. Then*

$$T(n, k) - T(n, k - 1) = \begin{cases} L_m, & m \geq 1, \\ 0, & m = 0, \\ -L_{-m}, & m \leq -1. \end{cases} \quad (13)$$

In particular, the adjacent-difference conjecture of A332709 holds on its full natural domain $4 \leq k \leq n$, $n \geq 2k - 3$.

Proof. First suppose $m \geq 1$. In Lemma 3.1 set $a = 2k - 7$ and $b = 2n - 2k + 1$. Its left side is the difference of the rook polynomials for $T(n, k)$ and $T(n, k - 1)$. Since $a + 2 = 2k - 5$ is odd and $b - a - 2 = 2m - 2$, it equals $-z^{2k-5}R_{2m-2}$.

From (4), $\Phi_N(z^d p) = (-1)^d \Phi_{N-d}(p)$ whenever the degrees permit it. Applying Φ_{n-1} therefore cancels the minus sign and gives $\Phi_m(R_{2m-2}) = L_m$, because $(n - 1) - (2k - 5) = m$.

If $m = 0$, the two entries form the central reflected pair in an even row and are equal by (6). If $m < 0$, reflection takes the difference to the negative of the left-half difference at $k' = n + 4 - k$, whose parameter is $-m > 0$. This proves all three cases. □

Remark 3.3. The $m = 0$ case is essential. A127548 has $L_0 = 1$, whereas the center difference is zero. Similarly, $k = 3$ has no predecessor column in the triangle. Neither case can be added to the positive-parameter formula by a zero-extension convention.

For later use, the left-half form is

$$A(n, s + 1) - A(n, s) = L_{n-2s-2} \quad (n \geq 2s + 3). \quad (14)$$

4 Concavity and all equality cases

Lemma 4.1. *For every $m \geq 1$, $L_{m+2} \geq L_m$. Equality holds exactly when $m = 1$. Also $L_m > 0$ for every $m \geq 1$ except $m = 2$.*

Proof. Let $\mathcal{S}_m$ be the permutations counted by L_m . For $\sigma \in \mathcal{S}_m$, define $\tau \in \mathcal{S}_{m+2}$ by

$$\tau(i) = \sigma(i) \ (i < m), \quad \tau(m) = m + 2, \quad \tau(m + 1) = \sigma(m), \quad \tau(m + 2) = m + 1.$$

These are distinct values and hence a permutation. The old restrictions remain valid. Row m receives $m + 2$, row $m + 1$ receives a value at most m , and the last row is unrestricted. The original permutation is recovered from the first $m - 1$ values and $\tau(m + 1)$, proving injectivity.

For $m \geq 2$, the shift by three on $m + 2 \geq 4$ letters is admissible and has value 1 at row m . Every image of the injection has value $m + 2$ there, so this shift lies outside the image. Hence the inequality is strict. Direct computation gives $L_1 = L_3 = 1$. Finally, the shift by two is admissible in S_m for $m \geq 3$, and the remaining cases are in (9). $\square$

Theorem 4.2 (Complete row shape). *Every row of T is positive, palindromic, unimodal and discretely concave:*

$$T(n, k + 1) - 2T(n, k) + T(n, k - 1) \leq 0 \quad (4 \leq k \leq n - 1). \quad (15)$$

Its maxima are exactly as follows.

- (i) *For odd $n \geq 3$, the unique maximum is at $k = (n + 3)/2$.*
- (ii) *For even $n \geq 6$, precisely $k = n/2, n/2 + 1, n/2 + 2, n/2 + 3$ are maxima.*
- (iii) *For $n = 4$, both existing entries are maxima.*

At internal columns, equality in (15) holds exactly at $k = (n + 1)/2, (n + 5)/2$ for odd n when those columns are internal, and at $k = n/2 + 1, n/2 + 2$ for even $n \geq 6$. Every odd row is strictly log-concave at every internal column. For even $n \geq 6$, equality in log-concavity holds exactly at the two interior columns of the four-entry plateau; for $n = 4$ there are no internal columns.

Proof. As k increases by one, the parameter $m = n - 2k + 4$ decreases by two. Lemma 4.1 and (13) show that the successive differences are nonincreasing on each side. At the junction they have the forms

$$\begin{aligned} n \text{ odd : } & \dots, 1, 1, -1, -1, \dots, \\ n \text{ even : } & \dots, 4, 0, 0, 0, -4, \dots, \end{aligned}$$

with indices outside the row omitted. Thus they are nonincreasing across the whole row, proving discrete concavity. They are nonnegative on the left and nonpositive on the right, proving unimodality.

For odd n , every step toward the center is positive, the final one being $L_1 = 1$. For even n , exactly the final left step L_2 , the reflected central step, and its right counterpart vanish. When $n \geq 6$ this is a four-entry plateau; all steps outside it are strict. The short rows are checked directly.

A second difference vanishes precisely when successive first differences agree. The sole positive-index equality $L_{m+2} = L_m$ is $L_3 = L_1$, giving the two stated odd positions. The even equality positions are the two internal points of the zero-difference run. No other equality is possible by strictness in Lemma 4.1.

For positive consecutive entries a, b, c with $2b \geq a + c$, one has

$$b^2 \geq \left(\frac{a + c}{2}\right)^2 \geq ac.$$

Equality requires $a = b = c$. It occurs precisely at the two internal points of the even plateau; the odd concavity equalities have unequal neighbors and therefore strict log-concavity. $\square$

The first rows illustrate the distinct parity patterns:

n	$(A(n, 1), \dots, A(n, n-2))$
3	(1)
4	(1, 1)
5	(4, 5, 4)
6	(20, 20, 20, 20)
7	(115, 116, 117, 116, 115)
8	(787, 791, 791, 791, 791, 787).

The theorem concerns a fixed- n row indexed by position. It is distinct in its variable from log-behavior of ordinary or straight ménage sequences indexed by n ; the limited literature check of that subject is discussed in Section 11.

5 The exact mean decomposition and a uniform tail

Define the relative row profile

$$\mathcal{R}(n, s) = \frac{(n-2)A(n, s)}{U_n}. \quad (16)$$

It has average one over the $n-2$ columns. The following identity is the bridge from exact differences to uniform asymptotics.

Theorem 5.1. *For every $n \geq 3$ and every allowed s ,*

$$A(n, s) = A(n, h) - \sum_{j=q}^{h-1} L_{n-2j-2}, \quad (17)$$

$$U_n = (n-2)A(n, h) - 2 \sum_{j=1}^{h-1} j L_{n-2j-2}, \quad (18)$$

$$\mathcal{R}(n, s) = 1 + \sum_{j=1}^{h-1} \left(2j - (n-2) \mathbf{1}_{\{q \leq j\}} \right) \frac{L_{n-2j-2}}{U_n}. \quad (19)$$

Empty sums apply to $n = 3, 4$.

Proof. Telescope (14) from the nearer endpoint side to column h , then apply reflection. This gives (17). For fixed $j \leq h-1$, the deficit term L_{n-2j-2} occurs in precisely the j leftmost and j rightmost columns. These sets are disjoint for either parity of n . Summing (17) proves (18); elimination of the central value proves (19). $\square$

Lemma 5.2. *For every integer $N \geq 2$, $\sum_{r=1}^N r! \leq 2N!$. The same upper bound therefore holds for a sum restricted to either parity.*

Proof. The case $N = 2$ is immediate. If the inequality holds at N , then the next sum is at most $2N! + (N+1)! \leq 2(N+1)!$, since $N+1 \geq 2$. $\square$

Theorem 5.3 (Uniform fixed-depth truncation). *For each fixed integer $J \geq 0$, as $n \rightarrow \infty$,*

$$\mathcal{R}(n, s) = 1 + \sum_{j=1}^J \left(2j - (n-2) \mathbf{1}_{\{q \leq j\}} \right) \frac{L_{n-2j-2}}{U_n} + O_J(n^{-2J-3}), \quad (20)$$

uniformly over all $1 \leq s \leq n-2$.

Proof. For large n we have $h > J$. Each omitted weight in (19) has absolute value at most n , independently of s . The omitted indices are positive integers of one parity no larger than $N = n - 2J - 4$. By (9) and Lemma 5.2, the omitted numerator is bounded in absolute value by $2nN!$. The classical equivalent $U_n \sim e^{-2}n!$, also proved by the expansion below, yields

$$\frac{2n(n - 2J - 4)!}{U_n} = O_J(n^{-2J-3}).$$

All constants are independent of the position. $\square$

This proof does not exchange an infinite asymptotic series with a growing sum. It first truncates an exact finite identity with a uniform factorial bound.

6 An all orders expansion engine

6.1 The classical circular expansion

Write $(v)_{\underline{k}} = v(v - 1) \cdots (v - k + 1)$ with $(v)_{\underline{0}} = 1$, and set

$$S_M(v) = \sum_{k=0}^M \frac{(-1)^k}{k!(v - 1)_{\underline{k}}}. \quad (21)$$

For every fixed $M \geq 0$, the classical Kaplansky–Riordan expansion is

$$U_m = e^{-2}m!\{S_M(m) + O_M(m^{-M-1})\}. \quad (22)$$

For completeness, a rigorous remainder follows immediately from the stronger Wyman–Moser nearest-integer formula [7, eq. (3.23), p. 475]:

$$U_m = \text{nint} \left(e^{-2}m! \sum_{k=0}^{m-1} \frac{(-1)^k}{k!(m - 1)_{\underline{k}}} \right), \quad m \geq 2. \quad (23)$$

This formula is used only for $m \geq 2$; the signed value $U_1 = -1$ is separately defined. If $a_{m,k} = 1/(k!(m - 1)_{\underline{k}})$, then for $0 \leq k \leq m - 2$,

$$\frac{a_{m,k+1}}{a_{m,k}} = \frac{1}{(k + 1)(m - 1 - k)} \leq \frac{1}{m - 1}.$$

For fixed M and sufficiently large m , the absolute tail after M is at most $a_{m,M+1}/(1 - 1/(m - 1)) = O_M(m^{-M-1})$. The normalized rounding error is at most $e^2/(2m!)$, smaller than every fixed inverse power. This proves (22) and its remainder, including $M = 0$. The expansion and nearest-integer refinement are prior results, not contributions of this report.

6.2 The auxiliary amplitude and coefficient formula

Set

$$E_M(v) = S_M(v) + 2 \sum_{r=1}^M \frac{S_M(v - r)}{(v)_{\underline{r}}}. \quad (24)$$

Then

$$L_m = e^{-2}m!\{E_M(m) + O_M(m^{-M-1})\}. \quad (25)$$

To see this, keep U_m and the M immediately preceding terms in (11). The remaining unsigned count terms have sum at most a constant times $(m - M - 1)!$ by $U_r \leq r!$ and Lemma 5.2; the finitely many exceptional signed terms cancel or are bounded. Dividing by $m!$ makes this $O_M(m^{-M-1})$. Substituting (22) in the finitely many retained terms proves (25). There is no derivative assumption on a remainder.

For fixed d and a desired finite order, the coefficients of L_{n-d}/U_n are consequently obtained by expanding

$$\frac{E_M(n-d)}{(n)_d S_M(n)}, \quad (26)$$

with M chosen sufficiently large. Each denominator tends to its nonzero leading value after the indicated power of n is factored out, so ordinary finite rational-series algebra is legitimate.

Theorem 6.1 (Position-uniform all-orders expansion). *For every fixed integer $K \geq 0$, there are explicitly computable functions $c_0(q), \dots, c_K(q)$ such that*

$$\mathcal{R}(n, s) = \sum_{r=0}^K c_r(q) n^{-r} + O_K(n^{-K-1}) \quad (27)$$

uniformly over all allowed positions. Each $c_r(q)$ depends on q through finitely many indicators $\mathbf{1}_{\{q \leq j\}}$. It stabilizes at its interior value whenever $2q + 1 > r$. For fixed q , the first difference from the interior expansion is $-n^{-2q-1}$.

Proof. Choose J with $2J + 3 \geq K + 1$ and apply Theorem 5.3. Expand each of the finitely many retained ratios by (26), far enough to allow multiplication by the weight of size $O(n)$. These weights depend on the position only through bounded indicators, so the resulting remainder remains uniform.

The j th ratio starts with n^{-2j-2} , since both normalized amplitudes have leading value one. Its central-mean contribution $2jL_{n-2j-2}/U_n$ begins at degree $2j + 2$, while its boundary subtraction $(n-2)L_{n-2j-2}/U_n$ begins at degree $2j + 1$ with coefficient one. For fixed q , the first present boundary subtraction is $j = q$, proving the last two assertions. $\square$

Here “interior” names the stabilized coefficient regime for a specified finite order; it does not require q/n to stay away from zero. If $q = q(n) \rightarrow \infty$, it reaches that regime at every fixed order.

6.3 Explicit coefficients and the fourth column

Let $x = 1/n$. Table 1 lists the coefficients at degrees 3 through 10 in $\mathcal{R}$; the constant is one and the degree-one and degree-two coefficients vanish. The supplied code computes these by exact fractions,

q	x^3	x^4	x^5	x^6	x^7	x^8	x^9	x^{10}
1	−1	−4	−12	−33	−90	−246	−622	−1067
2	0	2	15	75	311	1136	3629	9077
3	0	2	16	90	455	2268	11583	61099
4	0	2	16	90	456	2296	12051	67193
≥ 5	0	2	16	90	456	2296	12052	67238

Table 1: Exact rational-series coefficients, in this case all integers. Each row has a uniform $O(n^{-11})$ remainder in its indicated position regime.

keeping one extra degree before multiplication by $n - 2 = x^{-1} - 2$. This guard is needed to avoid

an incorrect last retained coefficient. It also checks the circular amplitude independently through the exact recurrence

$$(n-2)U_n = (n^2 - 2n)U_{n-1} + nU_{n-2} - 4(-1)^n \quad (n \geq 3), \quad (28)$$

and the line amplitude through $L_n - L_{n-1} = U_n + U_{n-1}$, both with exact rational series. Normalized by $e^{-2}n!$, the forcing in (28) is smaller than every inverse power, giving

$$u(n) = u(n-1) + \frac{u(n-2)}{(n-1)(n-2)}$$

as a formal coefficient relation.

[write] The table rederived, 7 October 2026. The write obtained the asymptotic series of $u(n) = U_n/(e^{-2}n!)$ from the recurrence (28) alone (order by order; it equals the expansion of S_M in (21)), the series of $L_m/(e^{-2}m!)$ from the line recurrence $L_m - L_{m-1} = U_m + U_{m-1}$, and the ratios L_{n-d}/U_n from these, and inserted them in (20): every entry of Table 1 is reproduced, with zero constant-free terms at degrees one and two. Against the exact counts, $n^{11}(\mathcal{R}(n, s) - \sum_{r \leq 10} c_r(q)n^{-r})$ is $2.5 \cdot 10^3$, $2.2 \cdot 10^3$, $2.1 \cdot 10^3$ for $q = 1$, between $4.6 \cdot 10^3$ and $6.0 \cdot 10^3$ for $q = 2$, and about $3.4\text{--}4.5 \cdot 10^5$ for $q = 3, 4, 5$, at $n = 60, 100, 140$ for $q = 1, \dots, 5$ (bounded, as the uniform $O(n^{-11})$ requires). This is an implementation of the same mathematics, not an independent proof.

For a fixed fourth column, $q = 4$ eventually, so Table 1 gives (2). It also shows uniformly for $q \geq 2$ that

$$\mathcal{R}(n, s) = 1 + 2n^{-4} + O(n^{-5}), \quad (29)$$

while at either endpoint

$$\mathcal{R}(n, 1) = 1 - n^{-3} - 4n^{-4} - 12n^{-5} + O(n^{-6}). \quad (30)$$

The posted A258667 leading equivalent $e^{-2}(n-1)!$ agrees with these results because $U_n/(n-2) \sim e^{-2}(n-1)!$. The linked Lean file's concluding assertion is an equivalence statement, not equality of every coefficient. It was read for scope; its compilation was not rerun for this report.

7 Exact and sharp distance from the uniform distribution

Choose a circular ménage permutation uniformly, and put $S = \pi(1) - 2$. Then $\Pr(S = s) = A(n, s)/U_n$. Let ν_n be the uniform law on $\{1, \dots, n-2\}$, and define total variation by half the sum of absolute probability differences. We first establish its exact form for every row.

Lemma 7.1 (Auxiliary growth and parity tails). *For $m \geq 3$,*

$$L_m = mL_{m-1} - U_{m-2}. \quad (31)$$

For every $m \geq 2$, $L_{m+1} \geq mL_m$ and $L_{m+2} \geq m(m+1)L_m$. For every $N \geq 1$,

$$\sum_{\substack{1 \leq j \leq N \\ j \equiv N \pmod{2}}} L_j \leq 2L_N. \quad (32)$$

Proof. The formal series $\mathcal{F}(w) = \sum_{r \geq 0} r!w^r$ satisfies $w^2\mathcal{F}'(w) = (1-w)\mathcal{F}(w) - 1$, by coefficient comparison. Put $g(z) = z/(1+z)^2$ and $\mathcal{L}(z) = \mathcal{F}(g(z))$. Since $g'(z) = (1-z)/(1+z)^3$, the chain rule for formal series yields

$$z^2\mathcal{L}'(z) = \frac{1-z}{1+z}(1+z+z^2)\mathcal{L}(z) - (1-z^2).$$

Using $\mathcal{U}(z) = (1 - z)\mathcal{L}(z)/(1 + z)$ and rearranging gives

$$\mathcal{L}(z) - 1 - z^2\mathcal{L}'(z) - z\mathcal{L}(z) = -z^2\mathcal{U}(z) - z^2.$$

Coefficients of degree $m \geq 3$ give (31). At $m = 2$ the extra term matters: $L_2 = 2L_1 - U_0 - 1 = 0$. At $m = 3$, the signed $U_1 = -1$ gives $L_3 = 1$.

For $m \geq 3$, append the fixed point $\sigma(m) = m$ to a circular ménage permutation on $[m - 1]$. The first $m - 2$ restrictions are unchanged. In row $m - 1$ the old restriction excludes $m - 1$ and 1, whereas the new one excludes $m - 1$ and m ; the appended value m was absent from the old image. Thus the result is counted by L_m , injectively, proving $U_{m-1} \leq L_m$. Apply (31) at $m + 1$ to obtain

$$L_{m+1} = (m + 1)L_m - U_{m-1} \geq mL_m \quad (m \geq 3).$$

The $m = 2$ case follows from $L_3 = 1$, $L_2 = 0$, and iteration gives the two-step inequality.

For $N = 1, 2, 3, 4$, the parity sums in (32) are respectively 1, 0, 2, 4, so the bound holds. For $N \geq 5$, induction and growth give

$$\sum_{\substack{j \leq N \\ j \equiv N \pmod{2}}} L_j \leq L_N + 2L_{N-2} \leq L_N \left(1 + \frac{2}{(N-2)(N-1)}\right) \leq 2L_N.$$

There is no division by L_2 ; the zero case was handled separately. $\square$

Theorem 7.2 (Exact sign pattern and total variation). *For $n = 3, 4, 6$ the row law is uniform. For $n = 5$ and every $n \geq 7$, precisely the two endpoints have mass below $1/(n - 2)$, and every interior entry has mass strictly above it. For every $n \geq 3$,*

$$\begin{aligned} d_{\text{TV}}(\mathcal{L}(S), \nu_n) &= 2 \left(\frac{1}{n-2} - \frac{A(n, 1)}{U_n} \right) \\ &= \frac{2}{n^4} + \frac{12}{n^5} + \frac{48}{n^6} + O(n^{-7}) \quad (n \rightarrow \infty). \end{aligned} \quad (33)$$

The distance has an expansion to every fixed order from the endpoint coefficients. Through degree ten, its next coefficients are 162, 504, 1500 and 4244.

Proof. Let $D(n, s) = (n - 2)A(n, s) - U_n$. For $n \geq 5$, the exact mean decomposition at $q = 2$ reads

$$D(n, 2) = 2L_{n-4} - \sum_{j=2}^{h-1} (n - 2 - 2j)L_{n-2j-2}. \quad (34)$$

The subtracted coefficients are nonnegative and at most $n - 6$. The indices run through one parity up to $n - 6$. For $n \geq 9$, Lemma 7.1 gives

$$\begin{aligned} D(n, 2) &\geq 2L_{n-4} - 2(n - 6)L_{n-6} \\ &\geq 2(n - 6)^2 L_{n-6} > 0, \end{aligned}$$

since $n - 6 \geq 3$. Formula (34) gives $D(5, 2) = 2$, $D(7, 2) = 1$ and $D(8, 2) = 8$. The remaining rows $n = 3, 4, 6$ are the constant rows already displayed.

Reflection and unimodality make every interior entry of a nonconstant row at least its second entry. All its interior deviations are therefore strictly positive. The deviations sum to zero, and the two endpoint deviations are equal, so both are strictly negative. Total variation equals the total negative mass, proving the exact identity. In the three constant rows both sides vanish; the singleton row $n = 3$ is handled separately rather than counting two distinct endpoints.

Finally, expand $2(1 - \mathcal{R}(n, 1))/(n - 2)$ using Table 1 and $(n - 2)^{-1} = n^{-1}(1 - 2/n)^{-1}$. This proves the coefficients and all-orders extension. $\square$

The exact sign classification requires no asymptotic onset. The numerical onset limitations for the inverse models below are separate and remain in force.

8 Fixed column inverse models and exact range recovery

8.1 Three inverse notions

Fix an integer $s \geq 1$, and define $a_n = A(n, s)$ for integers $n \geq s + 2$. It is useful to distinguish:

- (i) the inverse of an explicitly chosen smooth asymptotic model;
- (ii) recovery of n from an exact range value $y = a_n$;
- (iii) the threshold inverse $N_s(y) = \min\{n \geq s + 2 : a_n \geq y\}$ for a general real target.

They are different objects. In particular, an asymptotically tiny error in a real inverse does not permit a general-input ceiling formula, because an arbitrarily small error can cross an integer. These distinctions and the generic inverse machinery already occur in the repository's canonical and combinatorial transseries texts [11, 12].

Let $P_{s,M}(t) = \sum_{r=0}^M c_r(s)t^r$, the fixed-column relative polynomial from Theorem 6.1. Define the specified model

$$F_{s,M}(x) = \frac{e^{-2}\Gamma(x+1)}{x-2} S_M(x) P_{s,M}(1/x) \quad (35)$$

on a sufficiently large positive half-line. For $M = 0$, both last factors are one. The factors approaching one are positive there, and their logarithmic derivatives are $O(x^{-2})$. The digamma expansion gives

$$(\log F_{s,M})'(x) = \log x + O_{s,M}(x^{-1}). \quad (36)$$

Thus the model is eventually strictly increasing and unbounded.

Theorem 8.1 (Recovery at exact range values). *For fixed s, M , the unique nearby real solution x_n of $F_{s,M}(x_n) = a_n$ exists for all sufficiently large n , and*

$$x_n = n + O_{s,M}\left(\frac{n^{-M-1}}{\log n}\right). \quad (37)$$

In particular, rounding x_n to the nearest integer recovers n eventually.

Proof. The forward expansions give

$$a_n = F_{s,M}(n)\{1 + O_{s,M}(n^{-M-1})\}, \quad \log a_n - \log F_{s,M}(n) = O_{s,M}(n^{-M-1}).$$

By (36), the model's logarithm changes by order $\log n$ between n and either endpoint of $[n-1, n+1]$. These endpoint values bracket $\log a_n$ for large n . Continuity and strict monotonicity give the unique root there. The mean value theorem, with derivative bounded below by a positive constant times $\log n$, proves (37). Its error is eventually less than one half. $\square$

The onset in this theorem is existential. The model is not an exact or canonical interpolation of the integer sequence. The accompanying floating-point roots are diagnostics, not certified round-off or remainder bounds.

8.2 A Lambert W starting point

Stirling's formula and the first forward coefficients give, for each fixed s ,

$$\log a_n = n(\log n - 1) - \frac{1}{2} \log n + c + \frac{13}{12n} + O_s(n^{-2}), \quad c = \frac{1}{2} \log(2\pi) - 2. \quad (38)$$

Indeed $\log \Gamma(n+1)$ contributes $1/(12n)$, $-\log(n-2)$ contributes $2/n$, and $\log S_M(n)$ contributes $-1/n$ when expanded to adequate order. The fixed-column relative correction starts only at order n^{-3} or later.

For y large, let

$$L = \log y - c, \quad X = \frac{L}{W(L/e)}, \quad \ell = \log X, \quad (39)$$

where W is the positive real branch, so $X(\log X - 1) = L$. At exact range values $y = a_n$,

$$n = X + \frac{1}{2} - \frac{23}{24X\ell} + O_s\left(\frac{1}{X^2\ell}\right). \quad (40)$$

To prove this directly, substitute $n = X + \delta$ in the smooth main expression of (38). The term $\delta\ell - \frac{1}{2}\ell$ forces $\delta_0 = 1/2$. The residual at order $1/X$ is

$$\frac{\delta_0^2}{2} - \frac{\delta_0}{2} + \frac{13}{12} = \frac{1}{8} - \frac{1}{4} + \frac{13}{12} = \frac{23}{24}.$$

It is cancelled by $-23/(24X\ell)$. The remaining residual is $O(X^{-2})$, and division by a derivative of order ℓ proves (40). Higher orders follow by finite reversion of the explicit model with a correspondingly accurate forward expansion. This is a specialization of standard inversion, not a new Lambert- W technique.

For a precise comparison, choose $M \geq 4$, put $G_M(x) = e^{-2}\Gamma(x+1)S_M(x)/(x-2)$, and compare its inverse with that of $F_{4,M}$. Since $P_{4,M}(1/x) = 1 + 2x^{-4} + O(x^{-5})$, the mean value theorem gives, at a common large target,

$$F_{4,M}^{-1}(y) - G_M^{-1}(y) = -\frac{2}{X^4 \log X} \{1 + o(1)\}. \quad (41)$$

The positive fourth-column correction lowers the inverse. This compares the two specified smooth models, not two undefined interpolations.

9 A two candidate threshold enclosure

The smooth-model range theorem does not by itself solve the threshold problem. The following enclosure is the appropriate consequence when the remainder constants are retained.

Theorem 9.1. *Fix $s \geq 1$ and $M \geq 0$, and write $F = F_{s,M}$. There are constants $C > 0$ and an integer $N \geq s+2$ such that for every integer $n \geq N$,*

$$|\log a_n - \log F(n)| \leq Cn^{-M-1}, \quad (42)$$

and $F_{\pm}(x) = F(x) \exp(\pm Cx^{-M-1})$ are strictly increasing for $x \geq N$. If

$$y > F_+(N), \quad y > \max_{s+2 \leq n < N} a_n, \quad (43)$$

where the second restriction is vacuous for an empty prefix, define

$$x_+(y) = F_+^{-1}(y), \quad x_-(y) = F_-^{-1}(y).$$

Then the exact threshold satisfies

$$\lceil x_+(y) \rceil \leq N_s(y) \leq \lceil x_-(y) \rceil, \quad (44)$$

and

$$0 \leq x_-(y) - x_+(y) = O_{s,M} \left(\frac{X^{-M-1}}{\log X} \right), \quad (45)$$

where X is from (39). Thus, for sufficiently large y , the enclosure contains at most two consecutive integers. If it contains two, one exact comparison at the lower candidate determines the threshold.

Proof. Equation (42) follows from the forward relative remainder and positivity. Increase N so that F and both envelopes are increasing: their logarithmic derivatives are $\log x + O(x^{-1})$ by (36), since the extra derivative is $\mp C(M+1)x^{-M-2}$.

The sequence itself is eventually strictly increasing. To prove this without differentiating an unknown remainder, write $\log a_n = \log F(n) + r_n$ where $r_n = O(n^{-M-1})$. Then $r_{n+1} - r_n = O(n^{-M-1}) = O(n^{-1})$, and integration of (36) gives

$$\log a_{n+1} - \log a_n = \log n + O(n^{-1}) > 0$$

for large n . Increase N again if needed. The growth of F and the remainder also make a_n unbounded, so the threshold exists.

The envelope inequality is $F_-(n) \leq a_n \leq F_+(n)$ for $n \geq N$. Conditions (43) exclude earlier indices and ensure that both roots lie above N . If $a_n \geq y$, then $F_+(n) \geq y$ and $n \geq x_+(y)$, proving the lower bound. Conversely, the integer $n = \lceil x_-(y) \rceil$ satisfies $F_-(n) \geq y$ and hence $a_n \geq y$, proving the upper bound.

Both roots are asymptotic to X , by the leading logarithm $x(\log x - 1)$. At $x = x_+$,

$$\log F_-(x_+) = \log y - 2Cx_+^{-M-1}.$$

At x_- the same logarithm equals $\log y$. The mean value theorem and a logarithmic derivative of order $\log X$ prove (45). Eventually this width is less than one, so its ceiling endpoints are equal or consecutive. In the latter case, testing $a_\ell \geq y$ at the lower integer ℓ selects ℓ if true and $\ell + 1$ otherwise, by the enclosure. $\square$

Scope of certification. The constants C, N in this theorem exist by the proof, but no numerical certified values are supplied. The theorem is therefore not an executable certified threshold algorithm. No assertion is made that $N_s(y) = \lceil F^{-1}(y) \rceil$ for every sufficiently large input. A practical certified implementation would need explicit global bounds, finite-prefix handling and rigorous root enclosures; floating-point examples do not provide those ingredients.

Remark 9.2 ([write] The inverses and the transseries volumes, 7 October 2026). Compared statement by statement with the repository's transseries volume [13]; the volume's symbols are not used here.

1. *The growth.* $\log a_n = n(\log n - 1) - \frac{1}{2} \log n + c + O(1/n)$ contains $n \log n$, so it is not of the exponential-power form of `p0: def: model1`; no instance of the volume's model is claimed.

2. *The Lambert start.* With $w = \log X - 1$ the equation $X(\log X - 1) = L$ of (39) becomes $w + \log w = \log(L/e)$: the volume's dominant block `p0:thm:lambert-core` with unit slope and unit logarithmic coefficient, whose positive-coefficient branch gives exactly $X = L/W(L/e)$. An exact instance.
3. *The refined start and the model inverse.* In the chart of the logarithm of the unknown the core $x(\log x - 1)$ is a monomial-logarithmic datum (`plt:def:lw-monomial-log-datum`), but the terms $-\frac{1}{2} \log x + c + \frac{13}{12x}$ of (38) enter, relative to it, as exponentially small functions of that chart variable, outside the series the template requires; so the write has not shown (40), the model inverse of Theorem 8.1 or (41) to be instances of `plt:thm:lw-template`. They are proved directly.
4. *Range recovery and thresholds.* $N_s(y)$ is the volume's staircase (`p0:def:three-inverses(1)`) of the eventually increasing column; $F_{s,M}$ is not an interpolation of a_n (the source says so), so Theorem 8.1 is an analogue of `p0:thm:staircase(3)` and Theorem 9.1, with its envelopes $F_{\pm}$, an analogue of `p0:thm:staircase(2)`; neither is an instance. The companion volume's exact two-candidate rule (`t3:thm:two-candidate of Combinatorial_Transseries_Inverses.tex`) concerns another sequence; the write did not compare that volume further.

10 Reproducibility and implementation safeguards

The accompanying ZIP contains this L^AT_EX source, the PDF, a source ledger with version pins, deterministic exact code, optional high-precision diagnostics and a build script. It contains no copied source-paper PDFs, internal research notes or review reports.

The exact core uses only the Python standard library. Its public functions check types and domains, rejecting booleans, floating indices and invalid columns. In particular, the adjacent-difference function distinguishes the center from L_0 , and the circular count documents its signed small values. It exposes no routine advertised as a certified inverse.

The verification suite checks the path recurrence and polynomial identity, every row through $n = 100$, all first and second differences, complete concavity and log-concavity equality classifications, maxima, the mean decomposition, and the L/U relation. A separate subset-DP permanent computes the original allowed-position cofactors through $n = 14$. Exhaustive permutations check the line counts and the injection through $m = 8$. Exact fraction arithmetic reproduces the table, its order consistency and first boundary separation, the total-variation coefficients and the inverse coefficient residual. The all-row total-variation identity, its precise sign exceptions and the auxiliary growth and tail bounds are checked as well. A recurrence route cross-checks the forward amplitudes.

Deliberately false values and invalid API calls are negative controls: a wrong signed U_1 , a wrong center step, the interior coefficient substituted for the fourth-column coefficient, a wrong total-variation constant and invalid inputs must all be detected. Large computed integers can be serialized and parsed in bounded decimal chunks without changing the caller's Python digit limit; a 600-row example is tested with the limit set to 640. Checks use explicit exceptions rather than Python `assert`, so they remain active with `python -O`. The normal and optimized exact JSON receipts must agree byte for byte.

Numerical diagnostics use `mpmath` at 100 decimal digits and write a separate JSON file marked diagnostic only. They compare exact counts with truncated ratios and solve the explicit models at exact range targets. They check numerical convergence and implementation consistency; they are not the proof of any asymptotic remainder or certified inverse onset. The build creates a

deterministic ZIP by sorting members and fixing archive timestamps and file modes. Rebuilding the L^AT_EX PDF also fixes its date-dependent metadata through `SOURCE_DATE_EPOCH` and pdfL^AT_EX controls. The README specifies dependencies and commands.

[write] **The shipped files, 7 October 2026.** In ProveIt the PDF and `MANIFEST.sha256` are not shipped; the programs are `code/*.py` (the builder as `code/build.py`), the receipts `data/receipts-*.json`, and `requirements.txt` and `source_pins.json` are under `data/`. The builder expects the delivered layout and runs only in a re-extracted archive; the report README gives commands that rerun the exact suite and the diagnostics on a scratch copy.

11 Source scope and historical limitations

The source ledger is bounded to the material actually checked on 4 October 2026. It is not an exhaustive priority search.

- A332709’s internal entry displayed revision #33, 1 February 2021, with both conjectures. A site-wide footer date is not an entry revision. The full revision history was not audited.
- The full published Shevelev–Moses paper was read. Its path-rook factorization and fixed-couple formula are credited; its page 10 motivates the row-unimodality question.
- Wyman–Moser’s original paper, especially printed pages 473–475, supplies the nearest-integer remainder argument and explicitly credits the ordinary expansion to Kaplansky–Riordan. These results are classical.
- Kagey’s full 2023 arXiv version was read, including its prefix counting and Fibonacci rook-polynomial framework. No theorem for the present adjacent differences or row concavity was located there. The 2024 publisher version was identified, but its full text returned an access error and was not independently inspected.
- Gu and Zhao’s 2021 publisher abstract and first-page preview were accessible; the full paper was paywalled. Its abstract concerns log-behavior in the size variable for ordinary and straight ménage numbers. This is suggestively distinct from fixed-size row concavity, but full non-overlap with the inaccessible paper is not claimed.
- A258667’s displayed 2026 proof comment and complete linked Lean source were read for mathematical scope. They establish leading equivalence. The Lean proof was not rerun, and the present higher-order corrections do not contradict it.
- The repository’s canonical and combinatorial transseries source files were checked at pinned Git blobs. They already supply general inverse, range-recovery and threshold distinctions. Those methods are credited, not rebranded as new here.

Targeted identifier and keyword searches found no earlier resolution of the two displayed A332709 conjectures among the inspected sources. The mathematical proofs here stand independently of that bounded search. We claim the stated theorems, not global historical novelty or a complete clearance of all related literature.

Remark 11.1 ([write] The OEIS entries, 7 October 2026). The write read the entries in the internal format on 7 October 2026. A332709 is still at revision #33 (1 February 2021), by Peter Kagey (20 February 2020), with the two comments “Conjecture: Rows are unimodal (i.e., increasing, then decreasing).” and “Conjecture: $T(n,k) - T(n,k-1) = A127548(n-2k+4)$ for $n \geq 2k - 3$.”, the latter

signed by Kagey, 22 January 2021, and a b-file by Kagey for rows $3 \leq n \leq 52$. Theorems 3.2 and 4.2 prove both posted statements, the second on its natural domain $k \geq 4$ (at $k = 3$ the left side has no predecessor column). A258667, revision #79 (19 August 2026), by Vladimir Shevelev and Peter J. C. Moses (7 June 2015), states that “it is natural to conjecture that $a(n) \sim e^{(-2)^*n!/(n-2)^*(1 + \text{Sum}_{\{k \geq 1\}} (-1)^k/(k!(n-1)_k))}$ ”, and a comment of Ralf Stephan (30 June 2026) reports a proof by an autonomous AI agent in a linked Lean file, which establishes equivalence to $e^{-2}(n-1)!$. Read as a leading equivalence, which is what the Lean file formalizes (with the sum over $1 \leq k \leq n-1$), the conjecture holds, as the source says. *The write adds:* read as an asymptotic expansion in powers of $1/n$, it fails at the fourth relative order. Indeed, by (23) the displayed expression with the finite sum equals $(U_n + \vartheta_n)/(n-2)$ with $|\vartheta_n| \leq \frac{1}{2}$, so by (2) $a(n)$ divided by it is $1 + 2n^{-4} + 16n^{-5} + O(n^{-6})$, not $1 + O(n^{-K})$ for every K . Numerically, n^4 times the ratio minus one is 3.10, 2.46, 2.22, 2.10 at $n = 20, 40, 80, 160$. The other column entries (A258664–A258666, A258673) post no asymptotic formula. A127548 (#38, 20 September 2024) and A000179 are as the source describes them. *Checks.* The write’s counts equal the b-files named in Section 1.2. No OEIS edit is part of this work.

[write] **Correction at the independent check, 7 October 2026.** The sentence “The other column entries (A258664–A258666, A258673) post no asymptotic formula” is wrong. A258664 (#85), A258665 (#75), A258666 (#74) and A258673 (#71), all of 5 November 2025 and by Shevelev and Moses (7 June 2015), each carry the same two comments as A258667 (A258664 has no space after the sum’s subscript): Riordan’s expansion of U_n and “Therefore, it is natural to conjecture that $a(n) \sim e^{(-2)^*n!/(n-2)^*(1 + \text{Sum}_{\{k \geq 1\}} (-1)^k/(k!(n-1)_k))}$.” None of them records a proof. Their data are the columns $s = 1, 2, 3, 5$ (checked against $T(n, s+2)$ for all 24 posted terms of each). *The check adds:* each conjecture holds as a leading equivalence and fails as an asymptotic expansion. By (23), $a(n)$ divided by the displayed expression with the finite sum is $\mathcal{R}(n, s) U_n/(U_n + \vartheta_n)$ with $|\vartheta_n| \leq \frac{1}{2}$. By Theorem 6.1 with $q = s$ eventually and Table 1, this is $1 - n^{-3} - 4n^{-4} + O(n^{-5})$ for A258664, $1 + 2n^{-4} + 15n^{-5} + O(n^{-6})$ for A258665, and $1 + 2n^{-4} + 16n^{-5} + O(n^{-6})$ for A258666 and A258673. It tends to one, but it is not $1 + O(n^{-K})$ for every K ; the same holds with the sum truncated at any fixed order. At $n = 40, 80, 160, 320$, n^3 times the ratio minus one is $-1.108, -1.052, -1.025, -1.013$ for A258664, and n^4 times it is 2.427, 2.200, 2.097, 2.048 for A258665 and 2.464, 2.215, 2.104, 2.051 for A258666 and A258673.

12 Further questions

Several natural next steps remain separate from the results proved above.

1. Make the fixed-column inverse constants C, N explicit, control the finite prefix and combine them with rigorous interval roots. This would turn the threshold enclosure into a certified algorithm.
2. Investigate whether similarly explicit total-variation identities hold for other conditioned permutation classes; the exact two-endpoint mechanism here may fail in more complicated forbidden boards.
3. Extend the path-identity approach to several fixed couples or prescribed prefixes. General rook factorizations are available, but row-shape inequalities for the resulting multidimensional arrays need separate proofs.
4. Investigate stronger positivity or higher-difference properties. Ordinary discrete concavity alone does not imply arbitrary higher-order sign patterns.
5. Complete the literature check of the inaccessible final publisher texts before making any historical-priority assertion.

[write] **Added 7 October 2026, under Vladimir’s standing rule of 4 October 2026.** Every open question of the source is in this list. The write adds from the non-claims a rerun of the Lean proof of the A258667 leading equivalent (read here, not compiled). The two A332709 conjectures are proved; the stronger, expansion reading of the A258667 conjecture is refuted at relative order n^{-4} (Remark 11.1); no claim of the source was found false.

[write] **Independent check of the write, 7 October 2026.** An independent adversarial check of the write read the OEIS entries and the Lean file again and recomputed every number the write added, with its own code. It took $T(n, k)$ from the entry’s double-sum formula, not from the cofactor formula (5), and U_n from Touchard’s formula.

- The counts equal all 1275 terms of the A332709 b-file, the A000179 and A127548 b-files, and the posted terms of A258664–A258667 and A258673. Theorem 3.2, the complete classification of Theorem 4.2, and the identity and sign pattern of Theorem 7.2 hold for every $n \leq 120$.
- Table 1 and the coefficients of (33) were tested against exact counts at $n = 100, 200, 300$, at the columns $q = 1, \dots, 5$ and the central column. Every residual $n^{11}(\mathcal{R}(n, s) - \sum_{r \leq 10} c_r(q)n^{-r})$ is bounded and settling (2239, 2055, 1996 for $q = 1$; $4.27 \cdot 10^5$, $4.13 \cdot 10^5$, $4.08 \cdot 10^5$ in the interior), and so is the total-variation residual (10793, 10709, 10680).
- The Wyman–Moser rounding (23) holds for $2 \leq m \leq 300$. The ratio of $a(n)$ to the A258667 expression gives n^4 times ratio minus one equal to 3.1016, 2.4644, 2.2150, 2.1036, 2.0509 at $n = 20, 40, 80, 160, 320$, and $n(n^4(\text{ratio} - 1) - 2)$ tends to 16, as Remark 11.1 says. The Lean file at the current main branch has 28,220 bytes, and with one newline appended it has the SHA-256 the source pins. Its `target_theorem_0` is an `IsEquivalent` statement.

It also confirmed the provenance figures (475,232 bytes, 16 files, 591 lines, 16 pages, the 15-entry manifest), the byte identity of the 12 staged files, Remark 9.2 against the volumes’ statements, the label numbering (60 delivered labels unchanged, 63 in all) and the README listing. It found one error and one stale statement, both corrected by notes. Remark 11.1 said that A258664–A258666 and A258673 post no asymptotic formula. In fact each posts the A258667 conjecture, and the report settles all four in the same way (note after the remark). The provenance note’s statement on the pinned volume blob has gone stale (note after it).

References

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- [2] The OEIS Foundation, *A127548*, formal generating function and identities; inspected 4 October 2026. <https://oeis.org/A127548>.
- [3] The OEIS Foundation, *A000179*, ménage numbers and signed Touchard conventions. <https://oeis.org/A000179>.
- [4] The OEIS Foundation, *A258667*, fourth fixed-couple column, including the 30 June 2026 leading-equivalence proof comment; inspected 4 October 2026. <https://oeis.org/A258667>.
- [5] V. Shevelev and P. J. C. Moses, *Alice and Bob Go to Dinner: A Variation on Ménage*, *Integers* **16** (2016), A72. <https://emis.de/ft/19616>.
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- [7] M. Wyman and L. Moser, *On the problème des ménages*, *Canadian Journal of Mathematics* **10** (1958), 468–480. <https://doi.org/10.4153/CJM-1958-045-6>.

- [8] P. Kagey, *Ranking and Unranking Restricted Permutations*, arXiv:2210.17021v2, 9 November 2023. <https://arxiv.org/abs/2210.17021v2>. Published in Discrete Applied Mathematics **355** (2024), 247–261, <https://doi.org/10.1016/j.dam.2024.05.010>; final publisher full text not inspected.
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- [10] Google DeepMind, *AlphaProof Nexus A258667 Lean source*, concluding `target_theorem_0`. https://github.com/google-deepmind/alphaproof-nexus-results/blob/main/APNOutputs/OEIS/oeis_A258667_conjecture_0.lean. Source SHA-256 in the supplement.
- [11] V. Reshetnikov, *Transseries: the polynomial-logarithmic calculus, series reversal at infinity, and inversion*, ProveIt repository, canonical source. Git blob `26378ed517d4ed2e21add80faacfd8851f1fbd10`; inverse distinctions, range recovery and gamma inversion sections. <https://github.com/VladimirReshetnikov/ProveIt>. Exact path and SHA-256 in the supplement.
- [12] V. Reshetnikov, *Combinatorial Transseries and Their Inverses*, ProveIt repository source. Git blob `17647298fa31d61c061d556a3b1e451a228ae20e`; inverse notions and interpolation discussion. <https://github.com/VladimirReshetnikov/ProveIt>. Exact path and SHA-256 in the supplement.
- [13] ProveIt, *Transseries: the polynomial-logarithmic calculus, series reversal at infinity, and the inversion of rapidly growing functions*, Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex (the source’s [11]); Part 0 (`p0:def:model`, `p0:thm:lambert-core`, `p0:def:three-inverses`, `p0:thm:staircase`) and the chapter on the Lambert template (`plt:def:lw-monomial-log-datum`, `plt:thm:lw-template`). [Added in the write.]